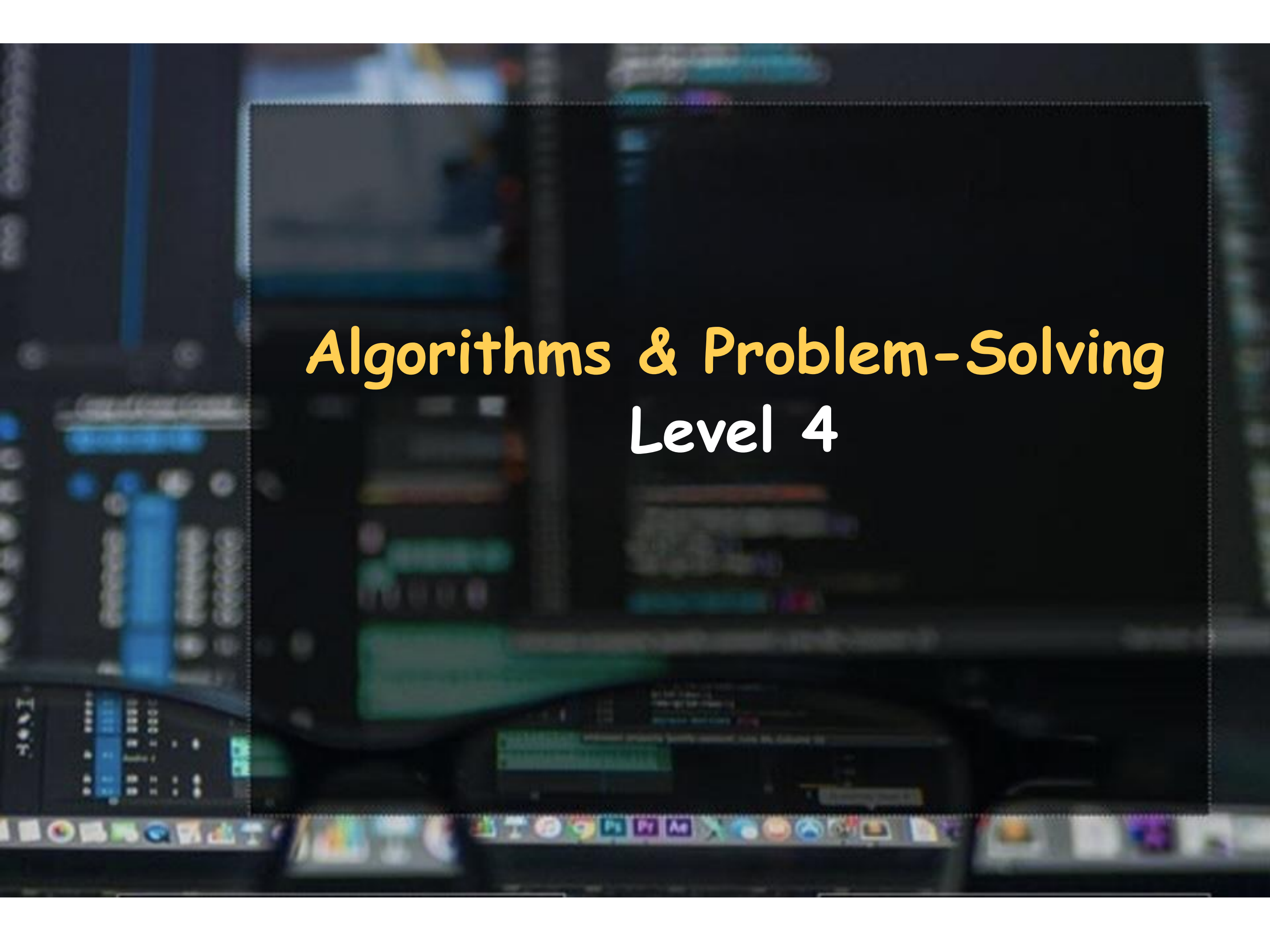




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, which adds a human element to the technical theme.

Algorithms & Problem Solving

Level 4

Validate Date

Definition

- sDate Struct: Represents the date and consists of:
 - Day
 - Month
 - Year
- IsLeapYear(year): Determines whether the year is a leap year or not.
 - A year is a leap year if:
 - It is divisible by 400, or it is divisible by 4 but not by 100.
- NumberOfDaysInAMonth(month, year): Returns the number of days in the month, taking into account the leap year for February.
- IsValidDate(date): Checks that the day and month are within the correct range, and that the day does not exceed the number of days allowed in that month.



Explanation:

- This program checks the validity of a date entered by the user (day, month, year) and verifies whether the date is correct in terms of the number of days in the month, the correctness of the month, and whether it is a leap year.

Procedures

- 1 The user enters the day, month, and year.
- 2 The number of days allowed for the given month is calculated.
- 3 The entered values are checked to be logical and correct.
- 4 The result is displayed: whether the date is correct or invalid.



Example:

Day = 35, Month = 1, Year = 2022
No, Date is NOT valid date

Day = 29, Month = 2, Year = 2022
No, Date is NOT valid date

Day = 29, Month = 2, Year = 2000
Yes, Date is a valid date.

Day = 31, Month = 4, Year = 2022
No, Date is NOT valid date

Explanation:

- If the month or day is outside the range (e.g., month 13 or day 0), the date will be considered invalid immediately.

Interpretation:

- The program compares the input day to the actual number of days in the month.
- The ``IsLeapYear()`` function is called to determine whether February has 28 or 29 days.

Summary

- ✓ The program thoroughly verifies the date using the following conditions:
 - Day between 1 and 31
 - Month between 1 and 12
 - Number of days does not exceed the allowed limit for the month
 - Correctly handles leap years

Thank you 😊